Positive Curvature of the Spectral Trace at the Critical Line

Ulrich Tehrani

Research Note · April 2026 · v1.3

Abstract

Building on the exact trace formula $\mathrm{Tr}(\widetilde{T}(\sigma)) = D_{\mathrm{SEL}} - O(\sigma)$ established in [5], we analyse the second-order curvature of the spectral trace at the critical line. The central quantity is the direct curvature sum $B = \sum_{k,p} e^{-\varepsilon^2\gamma_k^2}(\gamma_k \log p)^2 \cos(\gamma_k \log p)$, which satisfies the algebraic identity $O''(\frac{1}{2}) = -2B$, together with a Guinand–Weil decomposition of a smoothed prime–zero sum $\widetilde{Z}_p(\varepsilon)$ whose leading term is proved strictly negative. No hypothetical input is used: no Riemann Hypothesis, no Hilbert–Pólya postulate, no GUE conjecture.

What is proved. For every prime p and every $\varepsilon > 0$, the leading term of the Guinand–Weil decomposition satisfies $\mathrm{Main}_p(\varepsilon) < 0$ (Theorem 3.1), while the other-prime error is non-positive (Theorem 4.2) and the zero-sum truncation error at $N = 100$ is bounded by 3×10^{-61} relative to the main term (Theorem 4.3). The algebraic identity $O''(\frac{1}{2}) = -2B$ connecting curvature to bias is derived from the trace formula of [5] (Proposition 6.1). The decomposition $\widetilde{Z}_p(\varepsilon) = \mathrm{Main}_p + \Gamma_p + \mathrm{Const}_p + \mathrm{Err}_{p^k} + \mathrm{Err}_{\mathrm{other}}$ is the classical Guinand–Weil explicit formula applied to an even, Gaussian-damped test function; its derivation is recalled in § 2.

What is numerical. The gamma factor contribution is subleading, with $|\Gamma_p(\varepsilon)|/|\mathrm{Main}_p(\varepsilon)|$ numerically consistent with linear scaling in ε (Theorem 4.1). At the reference cutoff $\kappa = 53$, the constant-term ratio satisfies $r_p \in [0.75, 0.80]$ for every prime $p \leq 53$ (Theorem 5.1(a)). Direct grid evaluation on the tested values $\varepsilon \in \{0.020,\ 0.025,\ 0.030\}$ locates the sign-crossover in the interval $(0.020, 0.025)$, with $\mathrm{Re}\,\widetilde{Z}_p(\varepsilon) < 0$ at the tested $\varepsilon = 0.020$ for every prime $p \leq 53$ (Theorem 5.1(b)). The integrated bias is negative at reference parameters, $B_{\mathrm{int}}(0.05) = -42.21 < 0$ at $(\kappa, \varepsilon, N) = (53, 0.05, 100)$ (Theorem 6.2). At the same reference parameters the direct curvature sum equals $B = -19\,342.5 < 0$, which is equivalent to $O''(\frac{1}{2}) = +38\,685 > 0$ by the curvature–bias identity (Corollary 7.1).

What is conditional. If, in addition to the numerically established inequality $B < 0$ at the reference parameters, the stationarity condition $O'(\frac{1}{2}) = 0$ holds at the same parameters, then $\sigma = \frac{1}{2}$ is a strict local minimum of O and $\mathrm{Tr}(\widetilde{T}(\sigma))$ attains a strict local maximum at $\sigma = \frac{1}{2}$ at the reference parameters (Remark following Corollary 7.1). The conditional statement is clearly marked; its analytic hypothesis is the subject of Open Problem 8.1.

What remains open. Five questions separate the present work from a full curvature proof, each precisely named in § 8. Problem 8.1 (*Stationarity*) asks for an analytic bound $|O'(\frac{1}{2})| \ll W \cdot P$, where $W = \sum_k e^{-\varepsilon^2\gamma_k^2}$ and $P = \pi(\kappa)$. Problem 8.2 (*Asymptotic pointwise bias*) asks for an analytic proof of $\mathrm{Re}\,\widetilde{Z}_p(\varepsilon) < 0$ for every prime $p \leq \kappa$. Problem 8.3 (*Uniformity of positive curvature*) asks whether the inequalities $B < 0$ and $O''(\frac{1}{2}) > 0$ persist at parameter points (κ, ε, N) other than the reference values tested here. Problem 8.4 (*Integrated-to-direct transfer*) asks whether pointwise or integrated bias implies the negativ-

ity of B. Problem 8.5 (*Lagarias–Weil bridge*) asks whether the curvature statement transfers to the Weil-positivity framework of [7].

Structure. The formal core in §§ 2–7 is non-circular and proof-complete. No use of the Riemann Hypothesis, the GUE conjecture, the Montgomery pair correlation conjecture, or the Hilbert–Pólya postulate is made anywhere in §§ 2–7.

Notation and Conventions

Critical line. Throughout, the critical line means $\Re(s) = \frac{1}{2}$. The trace parameter $\sigma \in \mathbb{R}$ is evaluated at $\sigma = \frac{1}{2}$ unless stated otherwise.

Global parameters. The reference values are $\kappa = 53$ (prime cutoff, yielding the 16 active primes $p \leq \kappa$), $\varepsilon = 0.05$ (Gaussian regularisation parameter), and $N = 100$ (number of Riemann ordinates γ_k employed). These values are fixed throughout §§ 2–7 unless explicitly varied.

Main objects. The finite truncation

$$Z_p \;=\; Z_{p,\varepsilon,N} \;:=\; \sum_{k=1}^{N} e^{-\varepsilon^2\gamma_k^2}\, p^{i\gamma_k}$$

is the numerical object of [4, 5]. The associated Guinand–Weil object

$$\widetilde{Z}_p(\varepsilon) \;:=\; \sum_{\rho} h^{\mathrm{even}}_{p,\varepsilon}\left(\tfrac{\rho-1/2}{i}\right), \qquad h^{\mathrm{even}}_{p,\varepsilon}(u) = e^{-\varepsilon^2 u^2}\cos(u\log p),$$

sums over all non-trivial zeros of ζ. Both quantities are real-valued by the symmetric pairing $\rho \leftrightarrow \bar{\rho}$. $\Re Z_{p,\varepsilon,N}$ approximates $\Re\widetilde{Z}_p(\varepsilon)$ with truncation tail $R_{p,N}(\varepsilon)$; at $(\varepsilon, N) = (0.05, 100)$ this tail is negligible relative to $|\mathrm{Main}_p|$ by Theorem 4.3, hence all reported numerics are reproducible via $Z_{p,\varepsilon,N}$. The direct curvature sum

$$B \;:=\; \sum_{k,p} e^{-\varepsilon^2\gamma_k^2}\,(\gamma_k \log p)^2\, \cos(\gamma_k \log p)$$

enters the algebraic identity $O''(\frac{1}{2}) = -2B$ (Proposition 6.1); the integrated bias is $B_{\mathrm{int}}(\varepsilon) := \sum_p (\log p)^2\, \Re\widetilde{Z}_p(\varepsilon)$.

Not to be confused with. B (curvature sum, weight $(\gamma_k \log p)^2$) and B_{int} (integrated bias, weight $(\log p)^2$) differ in their weighting and in the object summed; both are negative at the reference parameters, but neither implies the other (Open Problem 8.4). Z_p (finite truncation, level N) and $\widetilde{Z}_p$ (Guinand–Weil object, sum over all non-trivial zeros) agree in real part for finite N but are formally distinct summations. The diagonal-energy constants $D \approx 9.471$ of [2] and $D_{\mathrm{SEL}} \approx 10.985$ of [5] share a name across the series but are defined in different operator contexts and must not be identified.

1 Introduction

1.1 Background and Motivation

This paper is the sixth in a series developing an operator-theoretic framework for studying the distribution of the non-trivial zeros of the Riemann zeta function $\zeta(s)$. The series is built around a finite-dimensional Hilbert-space model in which prime numbers mediate coupling between zeros, encoded in an explicit operator family.

Paper 1 [1] introduced the local curvature function $H_{\text{local}}(\sigma,\kappa) = \sum_{p\le\kappa} V_p(\sigma)$, identifying $\sigma = \frac{1}{2}$ as a phase boundary through the divergence $H_{\text{local}}(\frac{1}{2},\kappa) \sim 8(\log\kappa)^2$.

Paper 2 [2] connected this local curvature to the Weil explicit functional, constructing admissible test functions for which the curvature divergence becomes a structural feature of the Weil energy.

Paper 3 [3] made the geometric structure explicit: two finite-dimensional Hilbert spaces H_{str} (over primes) and H_{null} (over zeros) connected by a linear coupling map Φ, with loop operator $T = \Phi^* \circ \Phi$ and a first numerical proof of the energy asymmetry $\eta_{\text{orig}} > 0$.

Paper 4 [4] introduced the dual operator $\widetilde{T} = \Phi \circ \Phi^*$ on H_{null} — the Tehrani operator — and established a conditional proof of energy asymmetry under an equidistribution assumption on the prime log-phases.

Paper 5 [5] extended this to a one-parameter family $\widetilde{T}(\sigma) = \Phi(\sigma) \circ \Phi(\sigma)^*$ and proved the exact spectral trace identity

$$\mathrm{Tr}(\widetilde{T}(\sigma)) \;=\; D_{\text{SEL}} - O(\sigma), \qquad O(\sigma) \;=\; \tfrac{1}{2}\sum_{k,p} e^{-\varepsilon^2\gamma_k^2}\cos(2\sigma\gamma_k\log p),$$

where D_{SEL} is a σ-independent constant. Four independent numerical signatures concentrating at $\sigma = \frac{1}{2}$ were documented in [4, 5], and the curvature-bias quantity $B = \sum_p(\log p)^2\,\mathrm{Re}(Z_p)$ was introduced with the reference value $B = -19\,342.5$ at $(\kappa,\varepsilon,N) = (53, 0.05, 100)$.

The present paper addresses the analytic question left open by [5]: *is there a structural reason, independent of the Riemann Hypothesis, why $\sigma = \frac{1}{2}$ is distinguished by the trace?* We answer this in the affirmative at the level of second-order curvature, at the reference parameters. Twice-differentiating the trace identity at $\sigma = \frac{1}{2}$ yields the algebraic relation $O''(\frac{1}{2}) = -2B$, so that the sign of B determines the curvature of O at the critical line. At the reference parameters B is numerically negative, hence $O''(\frac{1}{2})$ is numerically positive. The paper establishes this curvature statement together with a Guinand–Weil decomposition of a related smoothed prime–zero sum, provides four numerical witnesses of the same qualitative phenomenon, and formulates five open problems constituting the completion path toward a theorem-level selection principle.

The operator $\widetilde{T}(\sigma)$ should be viewed neither as a Hilbert–Pólya spectral realisation in the sense of Connes [12] or Berry–Keating [13], nor as a de Branges structure-function criterion [14]; it is a finite-cutoff operator packaging Gaussian-smoothed explicit-formula data, closest in spirit to Hilbert-space reformulations of the explicit formula such as those of Burnol [15].

The heuristic labels "prime field" and "field strength" used in the series denote a finite-cutoff coupling picture in which primes mediate pairwise interactions between zero ordinates via explicit Gaussian-smoothed phasors; this should be distinguished from the periodic-orbit dictionary of Berry–Keating [13], the adelic trace-formula picture of Connes [12], and the Frobenius/cohomology picture of function-field zeta functions.

1.2 Main Results

The paper contains three main components.

The first is the curvature statement at the critical line. At the reference parameters $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, direct numerical evaluation gives $B = -19\,342.5 < 0$, which is equivalent via the algebraic identity of Proposition 6.1 to $O''(\frac{1}{2}) = -2B = +38\,685 > 0$ (Corollary 7.1). Independently, when combined with the stationarity hypothesis $O'(\frac{1}{2}) = 0$, this inequality implies that $\sigma = \frac{1}{2}$ is a strict local minimum of O at the reference parameters; the stationarity hypothesis is stated explicitly and left as Problem 8.1 of § 8.

The curvature identity $O''(\frac{1}{2}) = -2B$ is structurally analogous to the Li/Bombieri–Lagarias sign criteria [17, 18] in that it turns explicit-formula data into a sign condition, but it is not a derivative of ξ at $s = 1$ and not a Weil-positivity theorem; it is a local second-order statement in the auxiliary trace parameter σ at the symmetry point $\frac{1}{2}$.

The second is the Guinand–Weil bias analysis of §§ 2–5. The smoothed prime–zero sum $\widetilde{Z}_p(\varepsilon)$ is decomposed into five contributions, of which the leading one is proved negative (Theorem 3.1) and two further terms are proved non-positive or exponentially small (Theorems 4.2, 4.3). The gamma contribution is shown numerically to be subleading of order ε (Theorem 4.1). The constant-term ratio $r_p = \lim_{\varepsilon \to 0} |\mathrm{Const}_p(\varepsilon)|/|\mathrm{Main}_p(\varepsilon)|$ is numerically established to lie in $[0.75, 0.80]$ for every prime $p \leq 53$ (Theorem 5.1(a)), and a sign-crossover is localised in $(0.020,\ 0.025)$ (Theorem 5.1(b)). The integrated bias $B_{\mathrm{int}}(\varepsilon) = \sum_{p \leq \kappa} (\log p)^2 \operatorname{Re} \widetilde{Z}_p(\varepsilon)$ is numerically negative at the reference parameters, with $B_{\mathrm{int}}(0.05) = -42.21$ (Theorem 6.2).

The third is the structural separation made explicit in § 6. The quantities B, $B_{\mathrm{int}}(\varepsilon)$, and the pointwise sign of $\operatorname{Re} \widetilde{Z}_p(\varepsilon)$ are structurally distinct witnesses of the same qualitative phenomenon: B carries the $(\gamma_k \log p)^2$-weighting, B_{int} only the $(\log p)^2$-weighting, and the pointwise statement carries no weighting at all. All three are numerically negative at the reference parameters; the analytical relationship between them is left open as Problem 8.4. The five open problems of § 8 list the steps remaining, in order of what we consider tractability: Problem 8.1 (stationarity) is the most tractable and, together with the numerically established $B < 0$, would suffice to upgrade Corollary 7.1 to an unconditional strict local-minimum statement at the reference parameters.

1.3 Reader Contract

This paper *proves*, by direct computation and classical estimates of the Guinand–Weil type, that the main term is negative (Theorem 3.1), that the other-prime error is non-positive (Theorem 4.2), that the truncation error is negligible (Theorem 4.3), and the algebraic curvature–bias identity $O''(\frac{1}{2}) = -2B$ (Proposition 6.1). These are unconditional mathematical statements.

This paper *establishes numerically*, at the reference parameters $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, that the gamma contribution is subleading of order ε (Theorem 4.1), that $r_p \in [0.75, 0.80]$ for every prime $p \le 53$ and that the pointwise bias $\operatorname{Re} \widetilde{Z}_p < 0$ holds on the tested subgrid at $\varepsilon = 0.020$ (Theorem 5.1), that $B_{\text{int}}(0.05) = -42.21 < 0$ (Theorem 6.2), and that the positive-curvature statement $O''(\frac{1}{2}) = +38\,685 > 0$ holds at the reference parameters (Corollary 7.1). These statements are verified on a finite grid and are clearly labelled [numerical] in their theorem headings.

This paper *states conditionally* that $\sigma = \frac{1}{2}$ is a strict local minimum of O at the reference parameters, under the hypothesis $O'(\frac{1}{2}) = 0$ (Remark following Corollary 7.1). The hypothesis is the object of Problem 8.1, and no part of the unconditional claims above depends on it.

This paper *leaves open* five well-posed questions named in § 8: Problem 8.1 (Stationarity), Problem 8.2 (Asymptotic pointwise bias), Problem 8.3 (Uniformity of positive curvature), Problem 8.4 (Integrated-to-direct transfer), and Problem 8.5 (Lagarias–Weil bridge). The first four are self-contained within classical analytic number theory; the fifth names the remaining bridge to Weil positivity in the formulation of [7].

2 The Guinand–Weil Decomposition

Throughout, p denotes a rational prime, $\varepsilon > 0$ a smoothing parameter, and $\rho = \frac{1}{2} + i\gamma$ a non-trivial zero of the Riemann zeta function. We fix a cutoff $\kappa \ge 2$ and consider only primes $p \le \kappa$. For numerical reference we use $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, where N is the number of zeros used in the finite approximation; the first 16 primes lie below $\kappa = 53$.

Definition 2.1 (Smoothed zero sum)**.** *Let*

$$Z_p \;=\; Z_{p,\varepsilon,N} \;:=\; \sum_{k=1}^{N} e^{-\varepsilon^2\gamma_k^2}\, p^{i\gamma_k}, \qquad \operatorname{Re} Z_p \;=\; \sum_{k=1}^{N} e^{-\varepsilon^2\gamma_k^2} \cos(\gamma_k \log p).$$

This is the finite numerical object studied in [4, 5]. The associated object obtained from the Guinand–Weil explicit formula is

$$\widetilde{Z}_p(\varepsilon) \;:=\; \sum_{\rho} h^{\text{even}}_{p,\varepsilon}\!\left(\frac{\rho - \frac{1}{2}}{i}\right), \qquad h^{\text{even}}_{p,\varepsilon}(u) \;=\; e^{-\varepsilon^2 u^2} \cos(u \log p).$$

Since $h^{\text{even}}_{p,\varepsilon}$ is even and decays rapidly, $\widetilde{Z}_p(\varepsilon)$ is real-valued and absolutely convergent. Its Fourier transform is

$$\widehat{h}_{p,\varepsilon}(x) \;=\; \frac{\sqrt{\pi}}{2\varepsilon}\left[\exp\!\Big(-\tfrac{(\log p - 2\pi x)^2}{4\varepsilon^2}\Big) + \exp\!\Big(-\tfrac{(\log p + 2\pi x)^2}{4\varepsilon^2}\Big)\right],$$

which is strictly positive on $\mathbb{R}$ and concentrates at $\pm \log p/(2\pi)$ as $\varepsilon \to 0$.

Proposition 2.2 (Guinand–Weil explicit formula)**.** *For every prime p and every $\varepsilon > 0$,*

$$\widetilde{Z}_p(\varepsilon) \;=\; \mathrm{Main}_p(\varepsilon) + \Gamma_p(\varepsilon) + \mathrm{Const}_p(\varepsilon) + \mathrm{Err}_{p^k}(\varepsilon) + \mathrm{Err}_{\mathrm{other}}(\varepsilon),$$

where the five contributions are defined by the standard partition of the prime sum and the archimedean terms of the explicit formula. Explicitly,

$$\mathrm{Main}_p(\varepsilon) \;=\; -\frac{1}{2\pi}\cdot\frac{\log p}{\sqrt{p}}\Big[\widehat{h}_{p,\varepsilon}\Big(\tfrac{\log p}{2\pi}\Big) + \widehat{h}_{p,\varepsilon}\Big(-\tfrac{\log p}{2\pi}\Big)\Big],$$

$$\Gamma_p(\varepsilon) \;=\; \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-\varepsilon^2 t^2}\cos(t\log p)\,\mathrm{Re}\,\psi\big(\tfrac14 + \tfrac{it}{2}\big)\,dt,$$

$$\mathrm{Const}_p(\varepsilon) \;=\; \widetilde{Z}_p(\varepsilon) \;-\; \mathrm{Main}_p(\varepsilon) \;-\; \Gamma_p(\varepsilon) \;-\; \mathrm{Err}_{p^k}(\varepsilon) \;-\; \mathrm{Err}_{\mathrm{other}}(\varepsilon),$$

where ψ denotes the digamma function. The prime-power error collects the contribution from $n = p^k$ with $k \geq 2$, and the remaining-prime error from $n = q^k$ with $q \neq p$; both are standard sums of the form $\sum_n \Lambda(n)\widehat{h}_{p,\varepsilon}(\log n/(2\pi))/\sqrt{n}$.

Remark. *At the poles $s = 0, 1$ of ξ, the test function contributes the value $h^{\mathrm{even}}_{p,\varepsilon}(u)$ evaluated at $u = \pm i/2$. A direct computation gives*

$$h^{\mathrm{even}}_{p,\varepsilon}\big(\tfrac{i}{2}\big) + h^{\mathrm{even}}_{p,\varepsilon}\big(-\tfrac{i}{2}\big) \;=\; 2\,e^{\varepsilon^2/4}\cosh\Big(\tfrac{\log p}{2}\Big) \;=\; e^{\varepsilon^2/4}\Big(\sqrt{p} + \tfrac{1}{\sqrt{p}}\Big).$$

This is the hyperbolic cosine, not the circular cosine: the formal substitution $\cos(i\log p/2) = \cosh(\log p/2)$ *is essential. Replacing* cosh *by* cos *would yield a quantity that is negative for $p \geq 29$, which is analytically impossible since the left-hand side equals $\sqrt{p} + 1/\sqrt{p} > 2$ for every $p \geq 2$. The correct* cosh *form is used throughout what follows. See Edwards [11, §1.16 and §3.6] for the classical derivation of the pole contributions at $s = 0, 1$.*

The remainder of the paper studies the five terms of the decomposition in turn.

3 The Main Term

Theorem 3.1 (Main term negativity; proved)**.** *For every prime p and every $\varepsilon > 0$,*

$$\mathrm{Main}_p(\varepsilon) \;=\; -\frac{\log p}{2\sqrt{\pi}\,\varepsilon\,\sqrt{p}}\cdot\big(1 + O(e^{-(\log p)^2/\varepsilon^2})\big) \;<\; 0.$$

Proof. The Fourier transform $\widehat{h}_{p,\varepsilon}(x)$ is strictly positive on $\mathbb{R}$ and attains its maximum at $x = \pm\log p/(2\pi)$ with value $\sqrt{\pi}/\varepsilon$. The reflected value at $x = \mp\log p/(2\pi)$ is smaller by the factor $\exp(-(\log p)^2/\varepsilon^2)$, which is exponentially small: at $\varepsilon = 0.05$ and $p = 2$ it is below 10^{-82}, and decreases further for larger p. Collecting the dominant contribution,

$$\mathrm{Main}_p(\varepsilon) \;=\; -\frac{1}{2\pi}\cdot\frac{\log p}{\sqrt{p}}\cdot\frac{\sqrt{\pi}}{\varepsilon}\big(1 + O(e^{-(\log p)^2/\varepsilon^2})\big) \;=\; -\frac{\log p}{2\sqrt{\pi}\,\varepsilon\,\sqrt{p}}\big(1 + o(1)\big).$$

The sign is negative because all factors in the leading expression are positive and enter with a minus sign. $\square$

Corollary 3.2. *As $\varepsilon \to 0$, $|\mathrm{Main}_p(\varepsilon)| \sim (\log p)/(\varepsilon\sqrt{p})$, giving a leading-order scaling of $1/\varepsilon$. The overall constant is $C_{\mathrm{Main}} = -1/(2\sqrt{\pi}) \approx -0.2821$.*

At the reference parameter $\varepsilon = 0.05$ and for $2 \le p \le 53$, the main term takes values in the interval $[-4.15, -2.77]$, minimised at $p = 7$ and maximised at $p = 2$.

4 Error Terms

We now treat the three remaining contributions other than the constant term.

4.1 The gamma contribution

Theorem 4.1 (Gamma term is subleading; numerical)**.** *For every prime $p \le \kappa$ and every sufficiently small $\varepsilon > 0$, the ratio $|\Gamma_p(\varepsilon)|/|\mathrm{Main}_p(\varepsilon)|$ is numerically consistent with linear scaling in ε on the tested grid; a log–log fit over $\varepsilon \in [0.005, 0.1]$ yields slope* 1.001*. At $\varepsilon = 0.05$, the ratio is bounded above by* 0.172 *(attained at $p = 2$) and below by* 0.017 *(attained at $p = 53$); in particular $|\Gamma_p(0.05)| < |\mathrm{Main}_p(0.05)|$ for all* 16 *primes $p \le 53$.*

Sketch. The integrand in $\Gamma_p(\varepsilon)$ is uniformly integrable on $\mathbb{R}$ by the classical asymptotic $\mathrm{Re}\,\psi(\frac{1}{4}+it/2) = \log(1+|t|/2)+O(1)$ for large $|t|$ and the Gaussian weight $e^{-\varepsilon^2 t^2}$. Dominated convergence gives $\Gamma_p(\varepsilon) \to \Gamma_p(0)$ as $\varepsilon \to 0$, a finite limit. Since $|\mathrm{Main}_p(\varepsilon)|$ grows as $1/\varepsilon$, the ratio decays linearly. A log–log regression of the ratio against ε across $\varepsilon \in [0.005, 0.1]$ gives a slope of 1.001, consistent with the linear prediction to four significant figures. The analytic step that would upgrade this to a proved bound is the explicit control of $\Gamma_p(0)$ by a uniform constant in p, which we do not establish here; it is documented as part of Problem 8.2 in § 8. $\square$

4.2 The prime-power and other-prime errors

Theorem 4.2 (Other-prime error is non-positive; proved)**.**

$$\mathrm{Err}_{\mathrm{other}}(\varepsilon) \;\le\; 0 \qquad \textit{for every } \varepsilon > 0.$$

Proof. By the standard form of the explicit formula,

$$\mathrm{Err}_{\mathrm{other}}(\varepsilon) \;=\; -\frac{1}{2\pi} \sum_{\substack{q \le \kappa \\ q \ne p \\ q \text{ prime}}} \frac{\log q}{\sqrt{q}} \left[\widehat{h}_{p,\varepsilon}\left(\tfrac{\log q}{2\pi}\right) + \widehat{h}_{p,\varepsilon}\left(-\tfrac{\log q}{2\pi}\right)\right] + (\text{higher-prime-power terms}).$$

Each summand has $\log q > 0$, $\sqrt{q} > 0$, and $\widehat{h}_{p,\varepsilon}(\cdot) > 0$ on $\mathbb{R}$, with an overall minus sign. The higher-prime-power terms share the same sign structure with $\Lambda(q^k) = \log q$ and the same positive transform. Hence the full sum is non-positive. $\square$

Theorem 4.2 is a structural identity rather than an error bound: the contribution from primes $q \neq p$ actively reinforces the negativity of $\widetilde{Z}_p$.

Theorem 4.3 (Truncation error is negligible; proved)**.**

$$\frac{|R_{p,N}(\varepsilon)|}{|\mathrm{Main}_p(\varepsilon)|} \ \leq \ \exp(-\varepsilon^2\gamma_N^2) \cdot \frac{2\sqrt{\pi}\,\varepsilon\,\sqrt{p}}{\log p} \ \leq \ 3 \times 10^{-61}$$

at the reference parameters $(\varepsilon, N) = (0.05, 100)$ *for every prime* $p \leq 53$*, where* $R_{p,N}(\varepsilon) = \sum_{k>N} e^{-\varepsilon^2\gamma_k^2} \cos(\gamma_k \log p)$ *is the tail of the zero sum.*

Proof. The Gaussian weight $e^{-\varepsilon^2\gamma_k^2}$ majorises the tail. With $\gamma_{100} = 236.524$ and $\varepsilon = 0.05$ one has $\varepsilon^2\gamma_{100}^2 = 139.77$, so that $e^{-139.77} < 1.8 \times 10^{-61}$. The bound follows from the triangle inequality and the explicit lower bound on $|\mathrm{Main}_p|$ from § 3. $\square$

For every downstream claim in this paper, truncation at $N = 100$ is numerically equivalent to summing the full zero series.

5 The Constant Term and the Sign-Crossover Localisation

The constant term $\mathrm{Const}_p(\varepsilon)$ collects the pole residues of ξ, the $\log(2\pi)$ normalisation, and the finite remainder of the digamma contribution that is not captured in Γ_p. It is the only term of the decomposition that resists a short analytic treatment. We state a numerical result on its asymptotic ratio to the main term and, independently, a numerical result localising the sign-crossover of $\mathrm{Re}\,\widetilde{Z}_p$ on a finite ε-grid.

Theorem 5.1 (Asymptotic ratio bound and sign-crossover localisation; numerical)**.** *Numerically, the following hold.*

(a) *For every prime* $p \leq 53$*, the ratio* $|\mathrm{Const}_p(\varepsilon)|/|\mathrm{Main}_p(\varepsilon)|$ *appears to converge (numerical extrapolation) to a finite limit*

$$r_p \ := \ \lim_{\varepsilon \to 0} \frac{|\mathrm{Const}_p(\varepsilon)|}{|\mathrm{Main}_p(\varepsilon)|}$$

as $\varepsilon \to 0$*, with* $r_p \in [0.75, 0.80]$*.*

(b) *Direct evaluation on the tested values* $\varepsilon \in \{0.020, 0.025, 0.030\}$ *locates the sign-crossover in the interval* $(0.020, 0.025)$*. At the tested value* $\varepsilon = 0.020$*, one finds*

$$\mathrm{Re}\,\widetilde{Z}_p(\varepsilon) \ < \ 0 \qquad \textit{for every prime } p \leq 53.$$

Both statements are numerical and refer to the cutoff $\kappa = 53$*. The location of the sign-crossover and the sign at smaller tested values of* ε *are empirical findings on a finite grid; no continuous interval-persistence statement is asserted. The analytical question behind (a) is documented as Problem 8.2 of § 8.*

Discussion. The leading-order analysis of the pole-residue contribution at $s = 1$ of ξ uses the cosh-evaluation of the remark in § 2 and yields a formal expression of order ε^{-1}, matching the scaling of Main_p (see Edwards [11, §1.16 and §3.6]). The cancellation that determines the coefficient ratio is between $e^{\varepsilon^2/4}(\sqrt{p}+1/\sqrt{p})$ on the pole side and $-\log p/(2\sqrt{\pi}\,\varepsilon\,\sqrt{p})$ on the main-term side. Numerical evaluation of $\mathrm{Const}_p(\varepsilon)$ for $\varepsilon \in \{0.005, 0.010, 0.015, 0.020, 0.025, 0.030, 0.050\}$ across all primes $p \le 53$ shows that the ratio $|\mathrm{Const}_p(\varepsilon)|/|\mathrm{Main}_p(\varepsilon)|$ is a unimodal function of ε, with a maximum near $\varepsilon \approx 0.045$–$0.060$ and limiting value $r_p \in [0.75, 0.80]$ as $\varepsilon \to 0$. For $p \in \{37, 53\}$ this maximum narrowly exceeds unity (attaining 1.183 and 1.216 respectively), which produces two pointwise exceptions to the bias at $\varepsilon = 0.05$; see below. The sign-crossover localisation in (b) is obtained from the grid computation directly, not as an analytic consequence of (a).

Numerical evidence. At $\varepsilon = 0.05$ and $\kappa = 53$, the constant terms satisfy $\mathrm{Const}_p \in [2.44, 3.93]$ with mean 3.27 and standard deviation 0.46 (coefficient of variation 14%). Representative values are listed below.

p	$\mathrm{Main}_p(0.05)$	$\Gamma_p(0.05)$	$\mathrm{Const}_p(0.05)$	$r_p(\varepsilon \to 0)$
2	-2.77	-0.475	2.44	0.766
7	-4.15	-0.193	3.47	0.759
23	-3.69	-0.105	3.79	0.754
37	-3.35	-0.082	3.93	0.796
53	-3.08	-0.069	3.74	0.753

At $\varepsilon = 0.05$, the pointwise inequality $\mathrm{Re}\,\widetilde{Z}_p(0.05) < 0$ holds for 14 of the 16 primes $p \le 53$. The exceptions are $p = 37$ and $p = 53$, for which $\mathrm{Re}\,\widetilde{Z}_p(0.05) \approx +0.50$ and $+0.59$ respectively. This is *not* a failure of the underlying decomposition: it reflects the finite-ε maximum of the ratio $r(\varepsilon)$ near $\varepsilon \approx 0.05$, which narrowly exceeds unity for exactly these two primes. At the tested value $\varepsilon = 0.020$ the pointwise bias holds for all 16 primes without exception. At $\varepsilon = 0.025$ and $\varepsilon = 0.030$ the exceptions $p = 37, 53$ are present. The sign-crossover thus lies in the interval $(0.020,\ 0.025)$ as established by these three tested values. Whether the bias persists continuously throughout $\varepsilon \in (0, 0.020]$ is not established here.

No structural explanation for the exceptions $p = 37, 53$ at $\varepsilon = 0.05$ is currently available. The phenomenon appears to be a finite-ε, finite-κ manifestation of the observed unimodal peak of $r(\varepsilon)$ near $\varepsilon \approx 0.045$–$0.060$, which for exactly these two primes narrowly exceeds unity. Whether this reflects an underlying arithmetic structure or is a numerical coincidence at $\kappa = 53$ is not addressed here.

The observed negativity of $\mathrm{Re}\,\widetilde{Z}_p(\varepsilon)$ should be read neither as a prime race in the sense of Rubinstein–Sarnak [19] nor as a Landau–Gonek prime-power asymptotic; it is a different, prime-indexed sign statistic for Gaussian-smoothed explicit-formula evaluations.

Status. The statement $r_p < 1$ is verified numerically. A formal proof — which would close the asymptotic bias conjecture in its sharpest form — requires an explicit control of the residue-series expansion near the poles $s = 0, 1$ and is part of Problem 8.2 of § 8. The empirical sign-crossover localisation above, at $\kappa = 53$, is the sharpest statement that the present methods allow.

6 The Integrated Bias

The two expressions for B used below are the same quantity in two representations: the primal sum

$$B \;=\; \sum_{k,p} e^{-\varepsilon^2\gamma_k^2}\,(\gamma_k \log p)^2\,\cos(\gamma_k \log p)$$

and the dual sum $B = \sum_p (\log p)^2 \operatorname{Re} Z_p$. The equivalence between the two is Proposition 5.3 of [5] and is treated here as imported. All numerical values in this section refer to the same object.

We use the dual representation as a starting point. The curvature bias at $\sigma = \frac{1}{2}$ is encoded in the weighted sum

$$B \;:=\; \sum_{p \le \kappa} (\log p)^2 \operatorname{Re} Z_p,$$

which was introduced in [5] as $-\frac{1}{2}$ times the $\sigma = \frac{1}{2}$ evaluation of the second σ-derivative of the trace identity, via the identity $O''(\frac{1}{2}) = -2B$. Its structural decomposition is algebraic and does not depend on any unproved hypothesis.

Proposition 6.1 (Curvature–bias identity; proved)**.** *Let* $O(\sigma) = \frac{1}{2}\sum_{k,p} e^{-\varepsilon^2\gamma_k^2} \cos(2\sigma\gamma_k \log p)$ *and define*

$$B \;:=\; \sum_{k,p} e^{-\varepsilon^2\gamma_k^2}\,(\gamma_k \log p)^2\,\cos(\gamma_k \log p).$$

Then

$$O''(\tfrac{1}{2}) \;=\; -2\,B.$$

Proof. Direct differentiation gives

$$O'(\sigma) \;=\; -\sum_{k,p} e^{-\varepsilon^2\gamma_k^2}\,\gamma_k \log p\,\sin(2\sigma\gamma_k \log p),$$

$$O''(\sigma) \;=\; -2\sum_{k,p} e^{-\varepsilon^2\gamma_k^2}\,(\gamma_k \log p)^2\,\cos(2\sigma\gamma_k \log p).$$

Evaluation at $\sigma = \frac{1}{2}$ yields

$$O''(\tfrac{1}{2}) \;=\; -2\sum_{k,p} e^{-\varepsilon^2\gamma_k^2}\,(\gamma_k \log p)^2\,\cos(\gamma_k \log p) \;=\; -2\,B. \qquad \square$$

Numerically, at reference parameters $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, the direct curvature sum equals $B = -19\,342.5$, hence $O''(\frac{1}{2}) = -2B = +38\,685$. This value coincides with $\operatorname{Tr}(L)$, where $L = -d^2\widetilde{T}(\sigma)/d\sigma^2\big|_{\sigma=1/2}$ is the curvature operator of [5], and provides an independent closed-form cross-check for the identity above.

Remark. *The quantity B is defined as in [5, §5], and the identity $O''(\frac{1}{2}) = -2B$ follows* *from the explicit form of $O(\sigma)$ given there. The numerical value $B = -19\,342.5$ at $(\kappa, \varepsilon, N) = (53, 0.05, 100)$ is the one quoted throughout the series.*

Theorem 6.2 (Integrated bias is negative at reference parameters; numerical)**.** *At $(\kappa, \varepsilon, N) = (53, 0.05, 100)$,*

$$B_{\mathrm{int}}(0.05) \;:=\; \sum_{p \le 53} (\log p)^2 \operatorname{Re} \widetilde{Z}_p(0.05) \;=\; -42.21 \;<\; 0.$$

The direct curvature sum B of Proposition 6.1 is likewise strictly negative at the same parameters, with $B = -19\,342.5$, hence $O''(\frac{1}{2}) = -2B = +38\,685$.

Remark. *The values B and B_{int} refer to two* different *quantities, not to a single quantity in two forms. B is the direct curvature sum weighted by $(\gamma_k \log p)^2$, obtained by differentiating the trace identity and evaluating at $\sigma = \frac{1}{2}$; B_{int} is the integrated Guinand–Weil bias, weighted only by $(\log p)^2$ and summed over $\operatorname{Re} \widetilde{Z}_p$. The two differ by a reweighting through the γ_k^2-factors absent in the second. Both are numerically negative at the reference parameters, and the shared negative sign is a numerical finding rather than the consequence of a proved equivalence. Proposition 6.1 connects B — not B_{int} — to the curvature $O''(\frac{1}{2})$. Whether pointwise negativity of $\operatorname{Re} \widetilde{Z}_p(\varepsilon)$, or the integrated bias $B_{\mathrm{int}}(\varepsilon) < 0$, implies the corresponding negativity of B at the same parameters is an open question; it is formulated as Problem 8.4 of § 8.*

Remark. *Although $\operatorname{Re} \widetilde{Z}_p(0.05) > 0$ at $p = 37$ and $p = 53$, the $(\log p)^2$-weighted sum B_{int} remains decisively negative: the negative contributions from the remaining 14 primes dominate the positive contributions from these two. Concretely, $(\log 37)^2 \cdot 0.50 + (\log 53)^2 \cdot 0.59 \approx 6.52 + 9.30 = 15.82$, against a cumulative negative contribution of -58.03 from the other primes, yielding the net value -42.21. The integrated bias is robust against individual-prime deviations as long as these remain isolated.*

7 Positive Curvature at the Critical Line

We now record the curvature statement that follows from Theorem 6.2 and Proposition 6.1, and, separately, the conditional strict-local-extremum statement that follows once stationarity is added as a hypothesis.

Corollary 7.1 (Positive curvature at $\sigma = \frac{1}{2}$; numerical)**.** *At the reference parameters $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, the direct curvature sum B is numerically negative (Theorem 6.2), $B = -19\,342.5 < 0$. By Proposition 6.1, this is equivalent to*

$$O''(\tfrac{1}{2}) \;=\; -2B \;=\; +38\,685 \;>\; 0.$$

Proof. By Theorem 6.2, $B = -19\,342.5$. By Proposition 6.1, $O''(\frac{1}{2}) = -2B$. □

Remark. *If, in addition to $B < 0$ at the reference parameters (here numerically established), the stationarity condition $O'(\frac{1}{2}) = 0$ holds at the same parameters, then $\sigma = \frac{1}{2}$ is a strict local minimum of O, and the trace*

$$\operatorname{Tr} \widetilde{T}(\sigma) \;=\; D_{\mathrm{SEL}} - O(\sigma)$$

attains a strict local maximum at $\sigma = \frac{1}{2}$*.*

The vanishing of $O'(\frac{1}{2})$ *is left open and formulated as Problem 8.1 of § 8. Numerically,* $O'(\frac{1}{2}) = -2.475$ *is approximately* 11% *of the natural scale* $W \cdot P$*, consistent with the cancellation expected from the odd kernel* $\sin(\gamma_k \log p)$*, but no analytical bound on* $|O'(\frac{1}{2})|$ *is established in the present paper.*

Remark. *The paper [5] recorded four further numerical signatures concentrating at* $\sigma = \frac{1}{2}$*: a* $+47\%$ *trace spike relative to the* σ*-neighbourhood average, the dominance of the leading eigenvalue* μ_1*, the minimum of the spectral gap, and the joint maximum of* $\mathrm{Tr}(L)$ *and* $\lambda_{\max}(L)$*. Each of these is numerically robust and independent of the bias conditions established here. They constitute additional numerical signatures from [5] at the same reference parameters, but do not enter the proof of Corollary 7.1 or of the conditional remark above.*

Remark. *The curvature statement identifies* $\sigma = \frac{1}{2}$ *as a geometrically distinguished point of the spectral trace at the reference parameters. It is a structural ingredient in a possible route to Weil positivity in the Lagarias formulation [7] (Problem 8.5 of § 8), but does not by itself establish it. No implication concerning the Riemann Hypothesis is asserted or implied by Corollary 7.1, nor by the conditional remark above when the stationarity hypothesis is imposed.*

8 Open Problems

The present work reduces the question of curvature at the critical line to five well-posed subproblems. We name each and describe the tools we expect to be relevant. The problems are listed in order of what we consider tractability.

Open Problem 8.1 (Stationarity)**.** *Prove that* $|O'(\frac{1}{2})| \ll W \cdot P$*, where* $W = \sum_k e^{-\varepsilon^2\gamma_k^2}$ *and* $P = \pi(\kappa)$ *is the prime-counting function at the cutoff. The numerical value at reference parameters is* $O'(\frac{1}{2}) = -2.475$*, approximately* 11% *of* $W \cdot P$*, consistent with the cancellation expected from the odd kernel* $\sin(\gamma_k \log p)$*. The expected tools are: equidistribution of the sequence* $\{\gamma_k \log p \bmod 2\pi\}$ *under appropriate averaging; a discrete Weyl-type bound exploiting the independent behaviour of zero heights and prime logarithms. At the reference parameters,* $B < 0$ *is numerically established (Corollary 7.1); the remaining hypothesis of the conditional remark following it is therefore* $O'(\frac{1}{2}) = 0$*. A solution to this problem thus suffices on its own to discharge that remark at the reference parameters and yield an unconditional strict local-minimum statement for* O *at* $\sigma = \frac{1}{2}$ *at those parameters.*

Open Problem 8.2 (Asymptotic pointwise bias)**.** *Prove analytically that* $\mathrm{Re}\,\widetilde{Z}_p(\varepsilon) < 0$ *for every prime* $p \le \kappa$ *in some explicit range* $\varepsilon \in (0, \varepsilon_c)$*, together with the asymptotic ratio bound* $r_p < 1$ *for every prime* p*. At present both statements are verified numerically on the range* $p \le 53$*, with all observed values of* r_p *in* $[0.75, 0.80]$ *and with pointwise negativity established on the tested grid at* $\varepsilon = 0.020$*. Closing this gap would upgrade Theorems 5.1 to proved statements and, in combination with an analytic bound on the gamma contribution of Theorem 4.1, yield an analytic version of the asymptotic bias statement for* $\mathrm{Re}\,\widetilde{Z}_p(\varepsilon)$ *as* $\varepsilon \to 0$*. The expected tools are: explicit expansion of the pole residues of* ξ *at* $s = 0, 1$ *with the correct* $\cosh$*-evaluation of the test* *function; a Riemann–Lebesgue-type bound on the digamma contribution with explicit constant; a cancellation argument between the ε^{-1}-coefficients of* Main_p *and* Const_p*. The problem is self-contained within classical analytic number theory.*

Open Problem 8.3 (Uniformity of positive curvature)**.** *Determine whether the inequality $B(\varepsilon,\kappa) < 0$, and hence $O''(\frac{1}{2}) > 0$, persists at parameter points (κ, ε, N) other than the reference values $(53, 0.05, 100)$ tested here. Corollary 7.1 establishes the inequality at a single point of the parameter space. Whether it extends to a neighbourhood, to a larger range of cutoffs κ, or to a different choice of smoothing ε is a distinct question: the direct curvature sum depends non-trivially on the γ_k^2-weights and on the oscillatory phases $\gamma_k \log p$, and finite parameter changes could in principle reverse the sign. The expected tools are: a systematic numerical survey across a parameter grid; an asymptotic expansion of $B(\varepsilon,\kappa)$ for large κ; and, if possible, an analytic reduction via the primal-to-dual identity $B = \sum_p (\log p)^2 \operatorname{Re} Z_p$.*

Open Problem 8.4 (Integrated-to-direct transfer)**.** *Establish or disprove the implication*

$$\left(\operatorname{Re} \widetilde{Z}_p(\varepsilon) < 0 \text{ for all primes } p \le \kappa\right) \implies B(\varepsilon,\kappa) < 0$$

at fixed (ε,κ), where

$$B(\varepsilon,\kappa) \;:=\; \sum_{k,p} e^{-\varepsilon^2\gamma_k^2}\,(\gamma_k \log p)^2\,\cos(\gamma_k \log p).$$

All three quantities — the pointwise sign of $\operatorname{Re}\widetilde{Z}_p$, the integrated bias $B_{\mathrm{int}}(\varepsilon)$, and the direct curvature sum B — are numerically negative at the reference parameters $(\kappa,\varepsilon,N) = (53, 0.05, 100)$ (Theorems 5.1, 6.2, and Corollary 7.1), but no analytical implication among them is established. A proof of the displayed implication would extend Corollary 7.1 from a reference-parameter statement to a parametrised statement valid wherever the pointwise hypothesis can be verified. The expected tools are: the explicit prime-side representation of B, the uniform positivity of the Fourier transform $\widehat{h}_{p,\varepsilon}$, and a rearrangement argument distributing the γ_k^2-weights across the pointwise contributions. A natural intermediate step would be the analogous implication with B_{int} in place of B. The problem is self-contained within classical analytic number theory.

Open Problem 8.5 (Lagarias–Weil bridge)**.** *The present result introduces a finite-cutoff curvature observable for Gaussian-smoothed explicit-formula data; it should be compared with Bombieri's finite truncations of Weil's quadratic functional [16] and with the Li/Bombieri–Lagarias positivity framework [17, 18], but it is not yet a positivity theorem for the full Weil distribution.*

We conjecture that the positive-curvature statement $O''(\frac{1}{2}) > 0$, taken together with the stationarity condition $O'(\frac{1}{2}) = 0$, implies Weil positivity on the relevant class of smooth compactly-supported test functions in the formulation of Lagarias [7], and thereby implies the Riemann Hypothesis. The conjecture is stated here as a named target rather than as a proved implication; its resolution is beyond the methods of the present paper and requires either a direct positivity argument on the Weil distribution or an external route via Connes–Meyer-type operator positivity. Problems 8.1–8.4 are self-contained within classical analytic number theory; the present problem is structurally the principal outstanding question of the programme.

9 Non-Circularity

We document explicitly that the results of this paper do not rely on the objects they are directed toward.

No RH as input. At no point is the Riemann Hypothesis assumed. The ordinates γ_k are used as numerically specified inputs, and no property of their distribution requiring the Riemann Hypothesis is assumed or used.

No GUE, Montgomery, or Hilbert–Pólya hypothesis. The operator $\widetilde{T}$ and its σ-parametrised extension $\widetilde{T}(\sigma)$ are constructed explicitly from Gaussian-smoothed prime phasors in the previous papers of this series [3, 4, 5]. No random-matrix hypothesis, no pair-correlation conjecture, and no Hilbert–Pólya postulate enters the construction or the proofs.

$\widetilde{T}$ is not a Hilbert–Pólya operator. As established in [5], the eigenvalues of $\widetilde{T}$ decay as $\mu_j \sim C_T/\gamma_{k(j)} \to 0$, which is incompatible with the Hilbert–Pólya requirement that eigenvalues grow like the ordinates γ_k. The paper does not attempt a Hilbert–Pólya realisation.

The Guinand–Weil formula is classical. Proposition 2.2 is a classical identity valid for every even, sufficiently rapidly decaying test function; its derivation does not require the truth of the Riemann Hypothesis. Theorems 3.1, 4.2, and 4.3, together with the algebraic identity of Proposition 6.1, are proved unconditionally by direct computation.

Numerical results are clearly marked. Theorems 4.1, 5.1, and 6.2, and Corollary 7.1, rely on the numerical evaluation of finite sums whose truncation error is controlled by Theorem 4.3. Each such statement is labelled [numerical] in its heading.

Conditional statements are explicitly flagged. The conditional remark following Corollary 7.1 is stated conditionally on the stationarity hypothesis $O'(\frac{1}{2}) = 0$ (Problem 8.1) and adds no further unproved input beyond this explicit assumption. Problem 8.4 is named openly and is not used as a hypothesis anywhere in §§ 3–7.

γ_k as inputs, not conclusions. The ordinates appear as numerically specified inputs to the model. The claim $\mathrm{Re}(\rho_k) = \frac{1}{2}$ is not used anywhere; the reference point $\sigma_0 = \frac{1}{2}$ is chosen as the object of study, not assumed to be the unique zero location.

Acknowledgments

The author thanks the project's structured review and verification workflow for systematic error detection across draft versions, and external pre-review for successive rounds of feedback that shaped the final formulation of the results in this paper.

Call for Feedback

This paper is part of an ongoing open research initiative toward the Riemann Hypothesis. All papers in the series are freely available on Zenodo under CC BY 4.0, and the accompanying verification code is hosted on GitHub.

The author welcomes critical feedback, corrections, and collaboration: Zenodo (https://zenodo.org/search?q=Ulrich+Tehrani) and GitHub (https://github.com/utehrani/analysislab-nt).

Topics of particular interest include: analytical approaches to the asymptotic pointwise bias of Problem 8.2; the analytic stationarity bound of Problem 8.1; uniformity of the curvature inequality across a larger parameter range (Problem 8.3); the integrated-to-direct transfer of Problem 8.4; and, ultimately, the Lagarias–Weil bridge of Problem 8.5.

We particularly welcome expert comment on the following question: for a Gaussian-smoothed family g_ε with $g_\varepsilon \to \delta$ as $\varepsilon \to 0$, is it known whether the sign of $W(g_\varepsilon * g_\varepsilon^\vee)$ at finite ε carries arithmetic information, or is positivity only meaningful in the distributional limit?

References

[1] U. Tehrani, "A Curvature Decomposition of the Explicit Formula", Zenodo, https://doi.org/10.5281/zenodo.19025598, 2026.

[2] U. Tehrani, "From Local Curvature to the Weil Functional", Zenodo, https://doi.org/10.5281/zenodo.19106992, 2026.

[3] U. Tehrani, "A Finite-Cutoff Hilbert-Space Model for Prime–Zero Energy Structure", Zenodo, https://doi.org/10.5281/zenodo.19307989, 2026.

[4] U. Tehrani, "A Dual Operator for Prime–Zero Coupling and a Conditional Proof of Energy Asymmetry", Zenodo, https://doi.org/10.5281/zenodo.19364703, 2026.

[5] U. Tehrani, "Spectral Trace Formula and Smoothed Zero Sums: A Prime–Zero Duality Framework", Zenodo, https://doi.org/10.5281/zenodo.19508547, 2026.

[6] U. Tehrani, *Verification scripts for the series*, GitHub, https://github.com/utehrani/analysislab-nt.

[7] J.C. Lagarias, "On a positivity property of the Riemann ξ-function", *Acta Arithmetica* **89** (1999), 217–234.

[8] A.P. Guinand, "A summation formula in the theory of prime numbers", *Proc. London Math. Soc.* (2) **50** (1948), 107–119.

[9] A. Weil, "Sur les 'formules explicites' de la théorie des nombres premiers", *Comm. Sém. Math. Univ. Lund, Suppl.* (1952), 252–265.

[10] J. Hadamard, "Sur la distribution des zéros de la fonction $\zeta(s)$ et ses conséquences arithmétiques", *Bull. Soc. Math. France* **24** (1896), 199–220.

[11] H.M. Edwards, *Riemann's Zeta Function*, Academic Press, New York, 1974.

[12] A. Connes, "Trace Formula in Noncommutative Geometry and the Zeros of the Riemann Zeta Function", *Selecta Math. (N.S.)* **5**.1 (1999), 29–106.

[13] M.V. Berry and J.P. Keating, "The Riemann Zeros and Eigenvalue Asymptotics", *SIAM Review* **41**.2 (1999), 236–266.

[14] L. de Branges, "The Riemann Hypothesis for Hilbert Spaces of Entire Functions", *Bull. Amer. Math. Soc. (N.S.)* **15**.1 (1986), 1–17.

[15] J.-F. Burnol, "On Fourier and Zeta(s)", *Forum Math.* **16**.6 (2004), 789–840.

[16] E. Bombieri, "Remarks on Weil's Quadratic Functional in the Theory of Prime Numbers, I", *Rend. Lincei Mat. Appl.* **11**.3 (2000), 183–233.

[17] X.-J. Li, "The Positivity of a Sequence of Numbers and the Riemann Hypothesis", *J. Number Theory* **65**.2 (1997), 325–333.

[18] E. Bombieri and J.C. Lagarias, "Complements to Li's Criterion for the Riemann Hypothesis", *J. Number Theory* **77**.2 (1999), 274–287.

[19] M. Rubinstein and P. Sarnak, "Chebyshev's Bias", *Experimental Math.* **3**.3 (1994), 173–197.

Paper 6 v1.3 · April 2026
Series-paper bibitems: version numbers removed per template v1.4 (Concept-DOIs resolve to latest version)
Pauca sed matura